

INDUSTRY PROBLEM SOLVING TASK

CO4 AT2

ITA0502 – COMPUTER VISION

Submitted by

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1. Autonomous Vehicle Vision System Selection

The system requirements are:

- Accuracy must be **at least 90%**
- Latency must be **70 ms or less**
- Moderate computational resources are available.

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	95%	50 ms	Moderate
Deep Learning	98%	100 ms	High

Viola-Jones

Accuracy = **82%**, which is below the required 90%.

Although its latency is only 30 ms and computation is low, it **fails the accuracy constraint**.

Therefore, Viola-Jones cannot be selected.

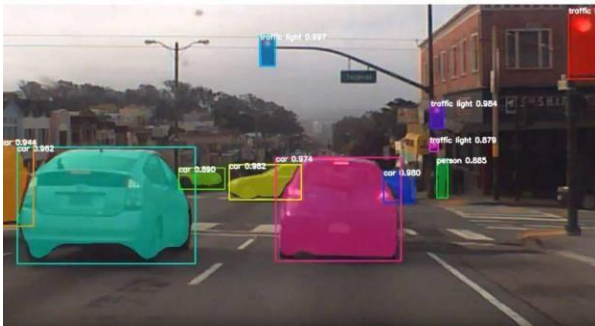


Figure 1: Computer Vision System for Autonomous Vehicle Detection

YOLO

Accuracy = **93%**, which satisfies the minimum 90% requirement.

Latency = **60 ms**, which is below the maximum acceptable 70 ms.

Computation = **Moderate**, which matches the available computational resources.

Therefore, YOLO satisfies **all three constraints**.

Deep Learning

Accuracy = **97%**, which satisfies the accuracy requirement.

However, latency = **90 ms**, which exceeds the maximum allowed 70 ms.

It also requires high computational resources, while the vehicle supports only moderate computation.

Therefore, Deep Learning cannot be selected.

Final Selection

Justification

YOLO is the only method that satisfies **accuracy, latency, and computational requirements simultaneously**.

- Accuracy: ✓
- Latency: ✓
- Computation: Moderate ✓

Therefore, **YOLO is the most suitable method for the autonomous vehicle vision system.**

2. Smart Factory Product Inspection Selection

The system requirements are:

- Accuracy must be **at least 95%**
- Processing time must be **50 ms or less**
- Hardware supports moderate computation.

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High

Traditional Image Analysis

Accuracy = **91%**, which is below the required 95%.

Although processing time is only 25 ms and computation is low, it fails the accuracy requirement.

Therefore, it cannot be selected.

YOLO

Accuracy = **96%**, which is above the required 95%.

Processing time = **45 ms**, which is below the maximum 50 ms.

Computation = **Moderate**, which matches the available hardware.

Therefore, YOLO satisfies **all constraints**.

Deep Learning

Accuracy = **98%**, which satisfies the accuracy requirement.

However, processing time = **75 ms**, which exceeds the maximum allowed 50 ms.

It also requires high computation, while only moderate computation is available.

Therefore, Deep Learning cannot be selected.

Final Selection

Justification

YOLO is the only method that satisfies all three requirements:

- Accuracy: ✓
- Processing time: ✓
- Computation: Moderate ✓

Therefore, **YOLO** is the most suitable solution for the smart factory product inspection system.



Figure 2: Computer Vision-Based Product Inspection in a Smart Factory